

# Targeted Unlearning via Discrepancy Minimization: Paper Replication and WMDP Application

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## Abstract

*This project investigates scalable approximate unlearning in Large Language Models (LLMs) under three increasingly realistic settings, focusing on suppressing targeted domain knowledge while preserving general-purpose reasoning ability. Building on prior work in discrepancy minimization, we develop an unsupervised unlearning pipeline that identifies domain-specific “anchor tokens” directly from model behavior using logit-ratio divergence between baseline and reinforced LoRA adapters, removing the need for manually curated entity substitutions. We additionally evaluate sparse selective unlearning across the Harry Potter series, The Iliad, and the King James Bible using semantic surprisal and emotional-volatility-based chunk selection. Results show substantial suppression of target-domain behavior, with Harry Potter anchor recall decreasing from 0.310 to 0.032 and WMDP-Bio and WMDP-Cyber accuracy approaching the 25% random-guess baseline, while ARC-Easy, BoolQ, and WinoGrande remain largely preserved. These findings suggest that approximate unlearning can scale beyond heavily supervised pipelines and suppress distributed hazardous or copyrighted knowledge with limited degradation of general model utility.*

## 1. Introduction/Background/Motivation

### 1.1. Objective

The objective of this project is to evaluate whether approximate machine unlearning can suppress targeted domain knowledge in large language models while preserving general-purpose reasoning ability. This matters because modern open-weight LLMs may be subject to lawsuits as they retain copyrighted training material [9], hazardous scientific knowledge [7], or offensive cybersecurity capabilities [10] long after pretraining.

Surface-level alignment methods such as RLHF can discourage the model from answering restricted prompts, but they do not necessarily remove the underlying knowledge.

If that knowledge remains latent in the model, adversarial prompting or jailbreaks may still recover it. A practical unlearning method should therefore make unwanted knowledge less accessible without requiring expensive retraining from scratch.

In this project, we study discrepancy-based approximate unlearning as a scalable approach to this problem. We extend prior work in three directions: an unsupervised pipeline that detects domain-specific anchor tokens from model behavior, sparse corpus selection methods based on semantic surprisal and emotional volatility, and supervised unlearning experiments on WMDP-Bio and WMDP-Cyber. Across these settings, we measure success by the tradeoff between targeted forgetting and utility preservation: an effective unlearned model should be less confident or less accurate on the target domain while maintaining strong performance on general benchmarks such as ARC, BoolQ, and WinoGrande.

### 1.2. Current Practice and Limits

Recent work on LLM unlearning has explored *approximate unlearning*, where a model is fine-tuned to suppress access to specific knowledge without retraining from scratch. A prominent example is Eldan and Russinovich’s *Who’s Harry Potter?* [3], which attempts to remove Harry Potter knowledge from Llama-2-7B. Their method first trains a reinforced model specialized on the target domain, then constructs sanitized training examples by replacing domain-specific entities with generic alternatives. During unlearning, the model is optimized to match a modified probability distribution that suppresses reinforced domain knowledge while preserving general language modeling behavior. The paper demonstrates that substantial forgetting can be achieved with relatively small amounts of GPU compute compared to full retraining.

However, the approach remains heavily supervised: developers must identify representative data, validate anchor terms, and construct substitutions manually. While feasible for a well-known corpus, this process becomes difficult to scale across specialized or rapidly changing domains where

subject-matter expertise may be unavailable.

Additionally, later work suggests that approximate unlearning may suppress model behavior without fully removing the underlying knowledge. Shostack [12] demonstrates that models claimed to be unlearned can still recover target-domain information under alternative prompting strategies. Similarly, To and Le [13] show that adversarial prompting techniques can reveal retained target knowledge even after successful benchmark unlearning results. These findings suggest that current approximate unlearning methods may reduce the accessibility of information without guaranteeing complete deletion from model parameters.

Another limitation of the original approach is that it focuses on unlearning a highly specific and self-contained corpus: the Harry Potter books. In many real-world settings, problematic knowledge is not confined to a single source, but instead appears across many documents, websites, and text distributions. Our work instead explores unlearning broader categories of knowledge, such as WMD-related content, where relevant information may be distributed across diverse sources and expressed in many different forms. This creates additional challenges for scalable unlearning, since there may not be a single canonical corpus or a well-defined set of target phrases.

Our work addresses these limitations by exploring a more unsupervised approximate unlearning pipeline that reduces reliance on manual domain expertise and human identification of target content.

### 1.3. Data

Our experiments utilized multiple data sources spanning copyrighted literary text, hazardous knowledge benchmarks, and automatically constructed domain corpora.

For the replication and unsupervised unlearning experiments, we constructed a Harry Potter corpus inspired by the setup of Eldan and Russinovich [3]. Raw text was extracted and cleaned from the Harry Potter book series, then tokenized and segmented into fixed-length training chunks. This entire corpus was used both to reinforce domain-specific knowledge through LoRA fine-tuning and to evaluate subsequent forgetting behavior.

For the novel application phase, we used the Weapons of Mass Destruction Proxy (WMDP) benchmark [6], specifically combining the `wmdp-bio` and `wmdp-cyber` subsets. These datasets contain expert-level multiple-choice questions designed to evaluate hazardous biological and cybersecurity knowledge while serving as a benchmark for targeted unlearning.

Unlike the original paper, which focused on a single copyrighted literary source, our project additionally explored broader hazardous knowledge domains derived from WMD-related material distributed across multiple sources and text styles. This provided a more realistic approxima-

tion of production unlearning settings, where undesirable knowledge may not exist inside a single canonical dataset.

To evaluate utility preservation, we additionally used standard downstream reasoning benchmarks including ARC-Challenge, BoolQ, and WinoGrande through the `lm-eval` evaluation framework. These benchmarks allowed us to measure whether targeted forgetting degraded the model’s performance as a general-purpose language model.

Following the recommendations of Gebru et al. [4], we document the provenance, intended use, and preprocessing pipeline of our datasets in order to clarify both the motivation for the data and the limitations of the resulting experiments. In particular, distinguishing between highly literary corpora and hazardous knowledge benchmarks is important for understanding how dataset structure affects approximate unlearning behavior.

## 2. Approach

### 2.1. Stage 1: Unsupervised Approximate Unlearning

We developed a unsupervised approximate unlearning pipeline based on distributional differences between a baseline model and a domain-reinforced variant. Rather than relying on manually curated translations or predefined target phrases, this approach attempts to automatically identify and suppress domain-specific knowledge directly from model behavior.

**Problem formulation.** Let  $\theta_{\text{base}}$  denote the frozen weights of `meta-llama/Meta-Llama-3-8B` and let  $\mathcal{D} = \{x^{(1)}, \dots, x^{(N)}\}$  be a target-domain corpus of  $N$  fixed-length token sequences. The goal is to learn a residual parameter update  $\phi_{\text{unlearn}}$ , implemented as a LoRA adapter, such that the adapted model  $\theta_{\text{unlearn}} = \theta_{\text{base}} + \phi_{\text{unlearn}}$  satisfies

$$p_{\theta_{\text{unlearn}}}(y | x) \approx p_{\theta_{\text{base}}}(y | x) \quad \forall x \in \mathcal{D},$$

while preserving  $p_{\theta_{\text{unlearn}}}(y | x) \approx p_{\theta_{\text{base}}}(y | x)$  for general-domain text. Crucially, the pipeline operates without manually labeled training examples, curated entity lists, or hand-crafted translation rules.

**Step 1: Reinforcement adapter.** We first train a LoRA adapter  $\phi_{\text{reinf}}$  ( $r=8$ ,  $\alpha=16$ , targets `q_proj/v_proj`) on  $\mathcal{D}$  by minimizing the standard causal language modelling objective:

$$\mathcal{L}_{\text{reinf}} = - \sum_{x \in \mathcal{D}} \sum_{i=1}^T \log p_{\theta_{\text{base}} + \phi_{\text{reinf}}}(x_i | x_{<i}).$$

Training uses AdamW ( $\text{lr}=2 \times 10^{-4}$ , effective batch size 16, 1 epoch). The resulting reinforced model  $\theta_{\text{reinf}} = \theta_{\text{base}} +$

$\phi_{\text{reinf}}$  over-fits to domain-specific continuations, making it a useful reference: positions where  $\theta_{\text{reinf}}$  and  $\theta_{\text{base}}$  diverge most strongly are precisely the positions most characteristic of the target domain.

**Step 2: Anchor token detection.** For each sequence  $x \in \mathcal{D}$  and each token position  $i$ , we compute the *logit ratio* — the factor by which reinforcement increased the model’s confidence on the true next token:

$$r_i = \frac{p_{\theta_{\text{reinf}}}(x_i | x_{<i})}{p_{\theta_{\text{base}}}(x_i | x_{<i}) + \varepsilon}, \quad \varepsilon = 10^{-9}.$$

Position  $i$  is designated an *anchor token* if  $r_i \geq \tau$ , where  $\tau = 2.0$ . The resulting binary mask is

$$\mathcal{A}(x) = \{i : r_i \geq \tau\}.$$

Across the 2,754 training chunks,  $|\mathcal{A}|$  averages 8.3% of positions per sequence, forming a sparse signal concentrated on domain-specific content.

**Step 3: Label replacement and unlearning objective.** A second LoRA adapter  $\phi_{\text{unlearn}}$  is trained on modified targets that steer the model away from domain-boosted predictions at anchor positions while leaving general syntax unchanged at non-anchor positions. Two complementary formulations were explored.

*Hard label replacement* (discrete approximation). For each position, the replacement label is:

$$\tilde{x}_i = \begin{cases} \arg \max_v p_{\theta_{\text{base}}}(v | x_{<i}) & i \in \mathcal{A}(x) \\ x_i & i \notin \mathcal{A}(x) \end{cases}$$

The unlearning loss is standard cross-entropy against these sparse replacement labels, with positions outside the sequence padded to the ignore index  $-100$ :

$$\mathcal{L}_{\text{hard}} = - \sum_{x \in \mathcal{D}} \sum_{i=1}^T \mathbf{1}[\tilde{x}_i \neq -100] \log p_{\theta_{\text{base}} + \phi_{\text{unlearn}}}(\tilde{x}_i | x_{<i}).$$

*Soft label replacement* (continuous target distribution). Rather than collapsing the baseline distribution to its argmax, we construct a full modified logit vector at each position using the ReLU-masked discrepancy formula from Eldan and Russinovich [3]:

$$v_{\text{generic}}(x_{<i}) = v_{\text{base}}(x_{<i}) - \alpha \cdot \text{ReLU}(v_{\text{reinf}}(x_{<i}) - v_{\text{base}}(x_{<i})),$$

where  $v_{\text{base}}$ ,  $v_{\text{reinf}}$  are the full vocabulary logit vectors from each model respectively and  $\alpha = 5.0$  controls suppression strength. The target distribution is  $p_{\text{target}}(\cdot | x_{<i}) = \text{softmax}(v_{\text{generic}}(x_{<i}))$ , and the loss minimizes KL divergence between it and the active unlearn adapter output:

$$\mathcal{L}_{\text{soft}} = \sum_{x \in \mathcal{D}} \sum_{i=1}^T \text{KL}(p_{\text{target}}(\cdot | x_{<i}) \parallel p_{\theta_{\text{base}} + \phi_{\text{unlearn}}}(\cdot | x_{<i})).$$

The ReLU is the critical element: it clamps the correction to be one-directional, activating only at positions where  $v_{\text{reinf}} > v_{\text{base}}$  (i.e., where reinforcement boosted a token’s logit). At positions where both models assign equal or lower probability under reinforcement, the ReLU evaluates to zero and  $v_{\text{generic}} = v_{\text{base}}$ , leaving the distribution unchanged. This asymmetry prevents the suppression signal from perturbing general syntax that was unaffected by domain fine-tuning. Both formulations are trained with AdamW (lr= $2 \times 10^{-5}$ , gradient accumulation steps 4) for 3 epochs.

**Relationship to discrepancy minimization.** The soft-label formulation is a generalization of the discrepancy minimization objective in [3]. The original work constructs  $p_{\text{target}}$  by substituting manually identified OOD proper nouns in the input text (e.g., “Harry”  $\rightarrow$  “a boy”) and computing the baseline model’s distribution on the modified input. Our formulation replaces this substitution step with the automatic anchor detection in Step 2, making the pipeline unsupervised: no named entity recognition, translation rules, or domain expertise are required. The logit-ratio threshold  $\tau$  plays the same role as the entity vocabulary in the original work, but is derived purely from the model’s own distributional shift under reinforcement.

## 2.2. Stage 2: Selective Text Extraction and Sparse Unlearning

We evaluated choosing specific parts of a corpus to be unlearned instead of requiring the model to unlearn an entire corpus. The general training strategy is identical to that in Stage 1. Our novel contribution demonstrates that effective unlearning can be achieved using only a sparse subset of the target corpus, extending the methodology to diverse, non-copyrighted historical texts.

**Data Selection and Sparsity** We selected 500 chunks each 300 words long. We selected text from 3 sources. A plain text extraction of the entire Harry Potter series, The Iliad, and the King James version of the Bible. We intentionally chose very old sources of text which have had ample time to culturally disseminate, potentially allowing knowledge about them to be learned from indirect sources.

This approximately 18% of The Bible, 13% of the Harry Potter Series and 80% of The Iliad. The Iliad was intentionally over sampled to demonstrate the effect of “dense” unlearning vs sparse unlearning.

**Random Sampling** As a baseline, we included randomly sampling chunks only to compare with the other two methods here.

**Pairwise Surprisal** We define surprisal as the average cosine distance in the embedding space.

$$d_{\cos}(u, v) = 1 - \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

We compute the mean pairwise surprisal for each embedding. We first L2-normalize each embedding

$$\hat{e}_i = \frac{e_i}{\|e_i\|_2}.$$

and compute a pairwise similarity matrix

$$S = \hat{E} \hat{E}^\top,$$

where row  $i$  of  $\hat{E}$  is  $\hat{e}_i$ . The distance matrix is

$$D = 1 - S.$$

For  $n$  chunks, chunk  $i$ 's pairwise surprisal score is computed as

$$s_i = \frac{1}{\max(n-1, 1)} \sum_{j=1}^n D_{ij}.$$

Since  $D_{ii} \approx 0$ , this is effectively the mean distance from chunk  $i$  to all other chunks. High-scoring chunks are semantic outliers in the document-level embedding geometry and were hypothesized to be more informative samples of a corpus.

**Pairwise Emotional Volatility** The emotional analysis uses the Hugging Face text-classification pipeline with the 'j-hartmann/emotion-english-distilroberta-base' model [5]. For each chunk  $c_i$ , the classifier returns scores over a fixed set of emotion labels. After sorting labels to keep dimensions aligned, the scores form an "emotion vector"

$$p_i = (p_{i1}, p_{i2}, \dots, p_{iK}),$$

where  $K$  is the number of emotion categories and  $p_{ik}$  is the model score for emotion  $k$  on chunk  $i$ . The dominant emotion is

$$\ell_i = \arg \max_k p_{ik}.$$

We again do a pairwise comparison. First, normalizing the emotion vectors and computing

$$\hat{p}_i = \frac{p_i}{\|p_i\|_2}, \quad D_{ij}^{(p)} = 1 - \hat{p}_i^\top \hat{p}_j,$$

then assigning

$$v_i = \frac{1}{\max(n-1, 1)} \sum_{j=1}^n D_{ij}^{(p)}.$$

The pairwise score identifies chunks whose emotion profile is unusual relative to the entire document, not merely surprising relative to local context. The motivation for this approach is a similar hypothesis that such substrings may be more informative for unlearning.

**Extreme Outlier Removal** In each corpus we removed the extreme outliers keeping only the IQR between  $q_{\text{low}} = Q_{0.5}(X)$  and  $q_{\text{high}} = Q_{99.5}(X)$  as we found samples outside this range often corresponded to non-narrative text such as copyright notices, tables of contents, etc.

It keeps only scores satisfying

$$q_{\text{low}} \leq x_i \leq q_{\text{high}}.$$

**Evaluated Models** For surprisal we evaluated all-MiniLM-L6-v2 [14], all-mpnet-base-v2 [15], BAAI/bge-base-en-v1.5 [16], and nomic-ai/nomic-embed-text-v1.5 [8]. The preliminary results showed that BAAI/bge-base-en-v1.5 outperformed other models, so we used this model in the final analysis.

### 2.3. Stage 3: Supervised Discrepancy Minimization on WMDP

We applied Targeted Unlearning via Discrepancy Minimization to meta-llama/Meta-Llama-3-8B-Instruct. The approach forces the model's probability distribution to match a generic, sanitized target distribution.

To generate this target, we first trained a Low-Rank Adaptation (LoRA) adapter to hyper-focus on the hazardous domain ( $v_{\text{reinforced}}$ ). We then built a Natural Language Toolkit (NLTK) Part-of-Speech (POS) tagging pipeline to extract Out-of-Distribution (OOD) proper nouns (e.g., "SpyCEP", "Stuxnet") and translated them into generic hyponyms (e.g., "a protein", "a software tool") to create a sanitized input ( $x_{\text{translated}}$ ).

The optimization minimizes the Kullback-Leibler (KL) Divergence via Cross-Entropy between the active model and the generic target:

$$P_{\text{generic}} = \text{softmax}(v_{\text{baseline}} - \alpha \text{ReLU}(v_{\text{reinforced}} - v_{\text{baseline}}))$$

Our initial unlearning loop encountered Gradient Dilution. Because only 158 out of 3,260 sequences contained terms rare enough to be translated, the remaining 95% of the dataset acted as an identity mapping ( $x_{\text{original}} = x_{\text{translated}}$ ). We initially attempted dataset pruning, but found it statistically ineffective; traversing only 5% of the dataset failed to sufficiently dent the thousands of domain concepts tested in the evaluation harness.

Instead, we solved Gradient Dilution mathematically. By introducing a Rectified Linear Unit (ReLU) activation mask

into the target probability constraint, we safely processed the entire unpruned dataset. Because the ReLU clamps non-hazardous identity distributions to zero, the aggressive  $\alpha$  penalty primarily activated when the reinforced model exhibited domain over-confidence. This allowed us to safely scale the learning rate and discrepancy penalty without inducing catastrophic forgetting of generic syntax.

### 3. Experiments and Results

#### 3.1. Unsupervised Unlearning on Harry Potter

We evaluated the unsupervised pipeline on a corpus constructed from the complete Harry Potter series (all seven novels), yielding 3,061 chunks of 512 tokens each (2,754 training, 307 held-out for evaluation). All metrics are derived directly from the corpus — no labeled questions or expected answers are required at any stage.

**Reinforcement phase.** A LoRA adapter ( $r=8, \alpha=16$ , targets `q.proj/v.proj`) was fine-tuned on the training split for one epoch using AdamW ( $\text{lr}=2 \times 10^{-4}$ , effective batch size 16). Cross-entropy loss decreased from 1.84 to approximately 1.71 over 170 optimiser steps, indicating that the adapter successfully reinforced domain-specific knowledge in the base model.

**Anchor token detection.** Using a logit-ratio threshold of 2.0, we scanned all 2,754 training chunks to identify positions where the reinforced model diverged meaningfully from the frozen baseline. Per-chunk anchor rates follow a roughly normal distribution centred at a mean of 0.083, indicating that approximately 8.3% of token positions per chunk are flagged as domain-specific anchors, distributed uniformly across the series with no strong concentration in any particular chapter or section. As a qualitative example, Chunk 1883 (anchor rate 0.245) corresponds to the Polyjuice Potion transformation scene and has 100 of 200 tokens flagged, illustrating that high-anchor chunks cluster around plot-critical, domain-specific narrative passages.

**Unlearning phase.** A second LoRA adapter was trained for 3 epochs ( $\text{lr}=2 \times 10^{-5}$ , gradient accumulation steps 4,  $\alpha=5.0$ ) using KL-divergence loss against the soft ReLU-masked target distribution. KL divergence loss decreased smoothly from approximately 1.4 at initialisation to a stable plateau near 0.4 by the end of epoch 3, with no signs of training instability or catastrophic forgetting of general syntax.

**Results.** Table 1 summarises the before/after comparison. Anchor recall fell from 0.310 to 0.032 — an 89.7% reduction — indicating that the unlearned model exhibits a

substantially reduced ability to predict highly reinforced domain-specific continuations. Held-out book perplexity rose by 65.4% (5.41 to 8.95), suggesting that the model is substantially more surprised by Harry Potter text after unlearning. Control perplexity increased by only 9.0% (12.94 to 14.10), demonstrating that the ReLU suppression mask suggesting that the ReLU suppression mask concentrated optimization pressure primarily on domain-boosted positions. General-capability benchmarks were largely preserved: ARC-Easy improved marginally (+1.2%), BoolQ declined by 1.2%, and WinoGrande by 3.9% — all within the expected variance of a 100-sample evaluation.

| Metric               | Base   | Unlearned | $\Delta$ |
|----------------------|--------|-----------|----------|
| Anchor Recall ↓      | 0.310  | 0.032     | −0.278   |
| Book Perplexity ↑    | 5.412  | 8.954     | +3.542   |
| Control Perplexity → | 12.939 | 14.100    | +1.161   |
| ARC-Easy →           | 0.820  | 0.830     | +0.010   |
| BoolQ →              | 0.820  | 0.810     | −0.010   |
| WinoGrande →         | 0.770  | 0.740     | −0.030   |

Table 1. Unsupervised unlearning results on the complete Harry Potter corpus. Target metrics (anchor recall, book perplexity) move in the expected direction while utility benchmarks remain within  $\pm 3.9\%$ .

#### 3.2. Selective Text Extraction and Sparse Unlearning Results

We next tested whether targeted unlearning can work using only a sparse subset of each corpus. For each source, we selected chunks using random sampling, semantic surprisal, or emotional volatility, while keeping the unlearning procedure fixed.

Table 2 shows that target perplexity increases after unlearning for every corpus and selection method. This indicates that the model becomes less confident on held-out target text. The effect is strong for the Bible corpus, where unlearning roughly 15% of the corpus raises perplexity from 1.190 to as high as 7.474. Harry Potter shows a smaller but consistent increase, while the Iliad shows the largest absolute increase (16.129 to 29.330), indicating that dense unlearning is likely still superior to sparse unlearning.

Surprisal-based selection gives the largest perplexity increase for all three corpora: +6.284 on Bible, +1.289 on Harry Potter, and +13.201 on Iliad. This suggests that semantic outlier chunks provide a slightly stronger forgetting signal than random or emotion-based selection. However, random selection remains competitive, indicating that the unlearning objective itself is effective once given enough target-domain text.

Performance was maintained on standardized LLM benchmarks ([2], [1], [11]) as shown in Table 3.

Although random selection was very competitive, the consistent superior performance of the Surprisal-based se-

lection suggests it may be possible to find, or even learn, a maximally efficient subset for unlearning.

| Dataset / Prompt Type | Base   | Unlearned | Absolute Change | % Change  |
|-----------------------|--------|-----------|-----------------|-----------|
| Bible – Emotion       | 1.190  | 6.986     | ↑ +5.796        | ↑ 487.06% |
| Bible – Random        | 1.190  | 7.237     | ↑ +6.047        | ↑ 508.15% |
| Bible – Surprisal     | 1.190  | 7.474     | ↑ +6.284        | ↑ 528.07% |
| HP – Emotion          | 3.249  | 4.474     | ↑ +1.225        | ↑ 37.70%  |
| HP – Random           | 3.249  | 4.407     | ↑ +1.158        | ↑ 35.64%  |
| HP – Surprisal        | 3.249  | 4.538     | ↑ +1.289        | ↑ 39.67%  |
| Iliad – Emotion       | 16.129 | 28.697    | ↑ +12.568       | ↑ 77.92%  |
| Iliad – Random        | 16.129 | 27.853    | ↑ +11.724       | ↑ 72.69%  |
| Iliad – Surprisal     | 16.129 | 29.330    | ↑ +13.201       | ↑ 81.85%  |

Table 2. Selective unlearning results comparing target perplexity before and after unlearning.

| Dataset           | ARC-Easy |           | BoolQ |           | WinoGrande |           |
|-------------------|----------|-----------|-------|-----------|------------|-----------|
|                   | Base     | Unlearned | Base  | Unlearned | Base       | Unlearned |
| Bible – Emotion   | 0.82     | 0.83      | 0.80  | 0.83      | 0.74       | 0.75      |
| Bible – Random    | 0.82     | 0.82      | 0.80  | 0.81      | 0.74       | 0.74      |
| Bible – Surprisal | 0.82     | 0.82      | 0.80  | 0.82      | 0.74       | 0.76      |
| HP – Emotion      | 0.82     | 0.83      | 0.80  | 0.79      | 0.74       | 0.71      |
| HP – Random       | 0.82     | 0.81      | 0.80  | 0.80      | 0.74       | 0.75      |
| HP – Surprisal    | 0.82     | 0.83      | 0.80  | 0.79      | 0.74       | 0.76      |
| Iliad – Emotion   | 0.82     | 0.81      | 0.80  | 0.83      | 0.74       | 0.77      |
| Iliad – Random    | 0.82     | 0.82      | 0.80  | 0.82      | 0.74       | 0.76      |
| Iliad – Surprisal | 0.82     | 0.82      | 0.80  | 0.81      | 0.74       | 0.78      |

Table 3. General benchmark accuracy comparison between the base and selectively unlearned models across ARC-Easy, BoolQ, and WinoGrande. Results indicate that selective unlearning preserves general downstream capabilities with only minor fluctuations in benchmark performance across datasets and prompt-selection strategies.

### 3.3. WMDP Application Results

Success is measured by calculating the performance delta between the baseline model ( $\theta_{baseline}$ ) and the unlearned model ( $\theta_{unlearned}$ ) across two axes:

1. **Targeted Forgetting:** Accuracy drop on hazardous knowledge ( $\Delta_{target} = Acc_{WMDP}(\theta_{unlearned}) - Acc_{WMDP}(\theta_{baseline}) \ll 0$ ).
2. **General Utility:** Accuracy retention on standard tasks ( $\Delta_{utility} = Acc_{general}(\theta_{unlearned}) - Acc_{general}(\theta_{baseline}) \approx 0$ ). We evaluated utility using ARC-Challenge, BoolQ, and WinoGrande.

Evaluating the final unlearned model over ten epochs reveals a successful decoupling of targeted domain knowledge from general reasoning capabilities. By leveraging the `torch.relu` constraint during discrepancy minimization, the model processed the full unpruned dataset without suffering from gradient dilution or catastrophic forgetting.

As detailed in Table 4, the target domain metrics ( $\Delta_{target}$ ) fell sharply within just the first three epochs.

WMDP-Bio accuracy dropped from 41% to 29%, and ultimately settled at 28% by epoch 10. Similarly, WMDP-Cyber dropped from 35% down to 29%. It is important to note that because the WMDP benchmarks evaluate using four-option multiple-choice questions, a 25% accuracy score effectively represents random guessing. This indicates the targeted domain knowledge was successfully and substantially suppressed hazardous-domain benchmark performance.

Crucially, the general utility metrics ( $\Delta_{utility}$ ) were heavily preserved throughout this targeted decay. The unlearned model retained its exact baseline scores on both BoolQ (82%) and WinoGrande (70%) by epoch 10, and ARC-Challenge experienced only a minor, acceptable variance (-4%). These results confirm that the mathematical unlearning mask substantially reduced accessible hazardous-domain behavior.

| Task Evaluation | Baseline | Epoch 1 | Epoch 2 | Epoch 3 | Epoch 10 |
|-----------------|----------|---------|---------|---------|----------|
| WMDP-Bio        | 0.4100   | 0.3200  | 0.2800  | 0.2900  | 0.2800   |
| WMDP-Cyber      | 0.3500   | 0.3200  | 0.2900  | 0.2900  | 0.2900   |
| ARC-Challenge   | 0.4600   | 0.4500  | 0.4200  | 0.4400  | 0.4200   |
| BoolQ           | 0.8200   | 0.7800  | 0.8300  | 0.8300  | 0.8200   |
| WinoGrande      | 0.7000   | 0.7300  | 0.7400  | 0.7400  | 0.7000   |

Table 4. Progression of model accuracy over the training schedule. Target metrics (WMDP) approach the 25% random-guessing baseline, while utility tasks successfully retain underlying capabilities.

## 4. Conclusion

This project demonstrates that approximate unlearning can substantially suppress targeted domain behavior while preserving much of a model’s general-purpose utility. By extending discrepancy minimization beyond manually supervised entity replacement toward unsupervised anchor-token detection and sparse corpus extraction, our results suggest that scalable unlearning may be possible even for distributed hazardous or copyrighted knowledge domains. However, the experiments also highlight important limitations of current approximate unlearning methods, particularly the possibility of residual knowledge recovery under adversarial prompting. Future work should investigate stronger robustness evaluations, automated red-teaming, adaptive anchor selection, and scaling behavior on larger and more heterogeneous corpora.

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| Team Member         | Contributed Aspects                                                          | Details                                                                                                                                                                                                                   |
|---------------------|------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| William Joyner      | WMDP Application & Architecture                                              | Engineered the NLTK translation pipeline, LoRA-based VRAM-constrained multi-adapter switching architecture, and dynamic KL-divergence training loop.                                                                      |
| Eric Regina         | Text Extraction, Contrastive Sentiment/Surprisal Analysis, Sparse Unlearning | Designed and implemented text extraction and contrastive analysis workflows, experimental design and analysis of multiple corpus.                                                                                         |
| Gunnar Andra-Thomas | Unsupervised Unlearning Pipeline Implementation                              | Designed and implemented the end-to-end unsupervised unlearning pipeline, including corpus construction, LoRA-based reinforcement training, token-level anchor detection, and selective label replacement for unlearning. |

Table 5. Contributions of team members.

## 5. Appendix

### 5.1. Code Repository

Our code is available here: <https://github.com/ericreg/deep-learning-final-project>

The repository is structured as a modular experimental framework for studying approximate machine unlearning in large language models. It contains separate components for data preprocessing, model training, selective unlearning, and downstream evaluation.

For training with 24GB of VRAM on a single NVIDIA RTX 3090, caching the dense 128,256-dimension vocabulary logits for the entire dataset was intractable. We engineered a dynamic multi-adapter pipeline that swaps the frozen reinforced adapter and the active unlearning adapter into memory during the forward pass to compute  $P_{generic}$  on the fly.

The main pipeline consists of:

- Corpus construction and preprocessing for retain/forget datasets,
- LoRA-based fine-tuning of pretrained transformer models,
- Selective unlearning methods using discrepancy analysis and token-level anchor detection,
- Benchmark evaluation for forgetting effectiveness and utility preservation.

The implementation is built primarily on the Hugging Face ecosystem and uses:

- PyTorch for training and optimization,
- Transformers for pretrained language models and tokenization,
- PEFT for LoRA adapters,
- Datasets for dataset management,
- and standard scientific libraries such as NumPy and Pandas.

The repository also includes scripts for automated experimentation, prompt-based probing, and benchmark testing on datasets such as ARC-Easy, BoolQ, and Winogrande to evaluate the tradeoff between successful forgetting and preservation of general-purpose model capabilities.

### 5.2. Plots

**Figure 1 — Unlearning Results: Base Model vs. Unlearned Model**

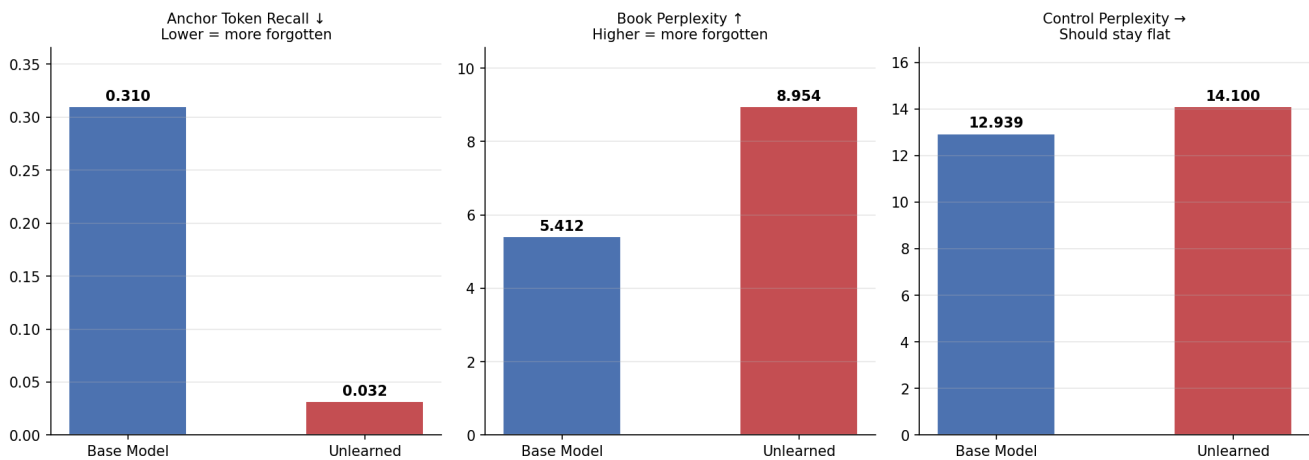


Figure 1. Dense unlearning results for Harry Potter.

Figure 1 — Selective Unlearning Results: Base vs. Unlearned

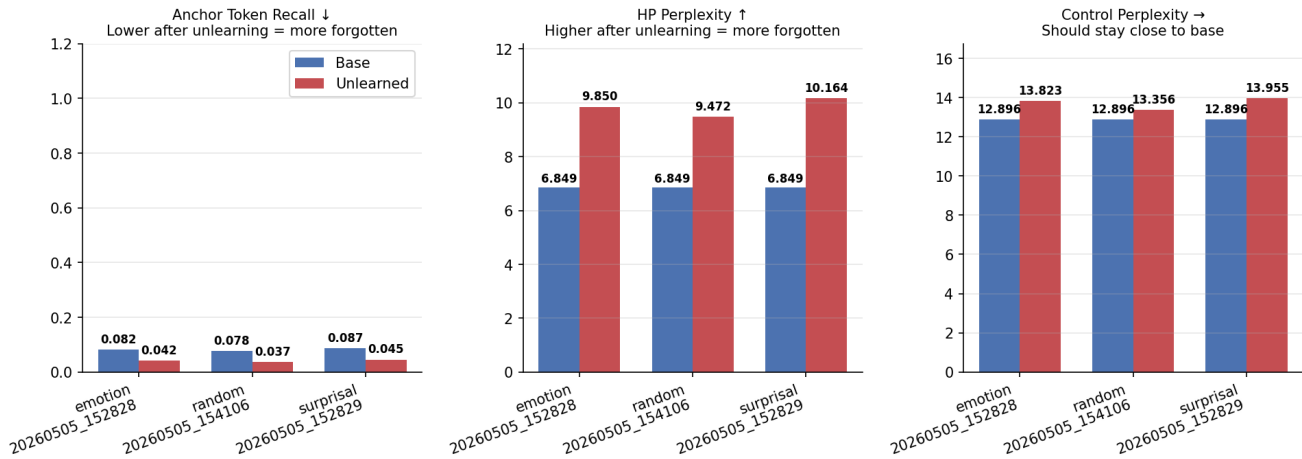


Figure 2. Dense unlearning results for Harry Potter.

Figure 3b — Unlearning Loss ( $\alpha=5.0$ ,  $lr=2e-05$ )

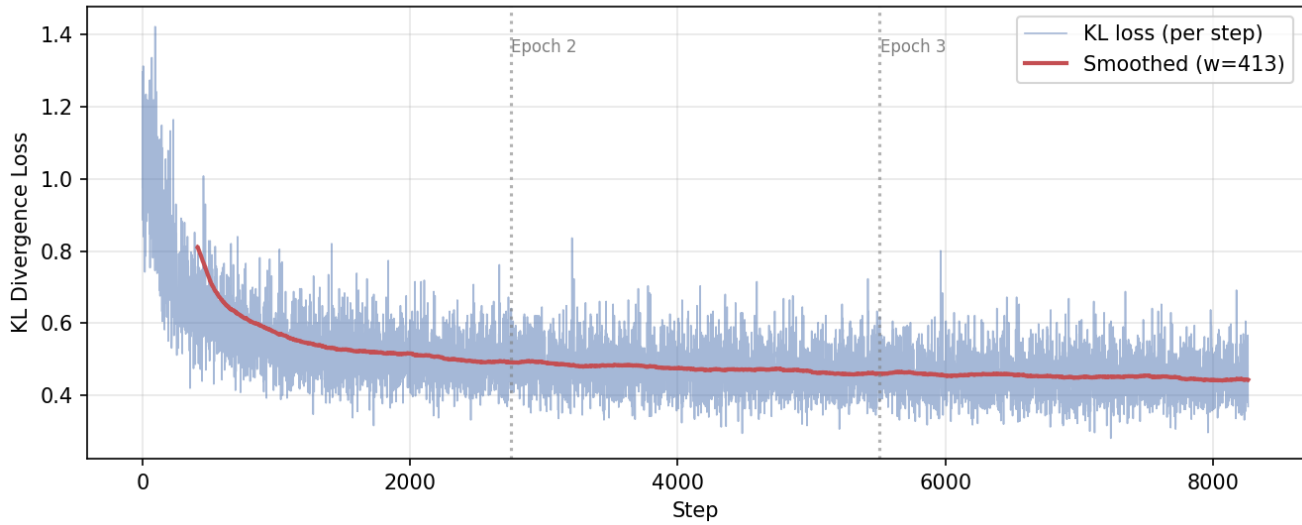


Figure 3. Training loss curve for dense unlearning on Harry Potter

Anchor Token Analysis | 2754 training chunks | threshold=2.0

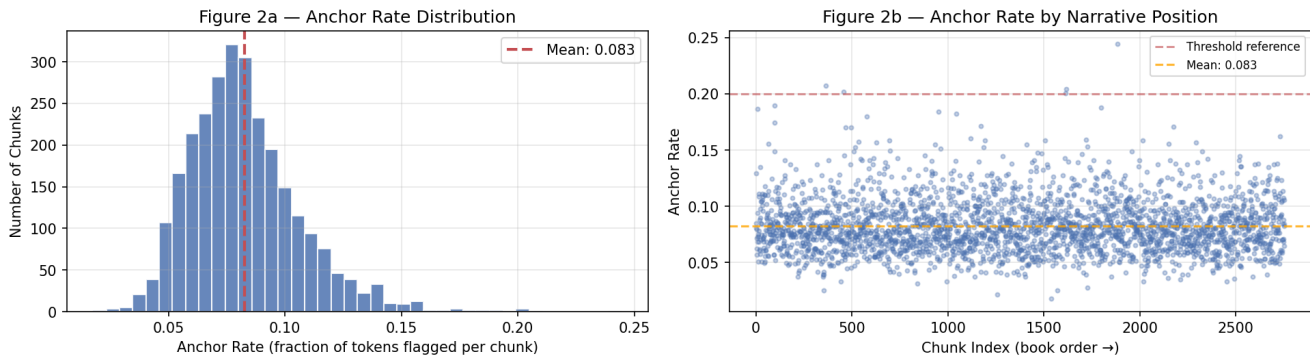


Figure 4. Anchor Rate Distribution for Dense Unlearning.

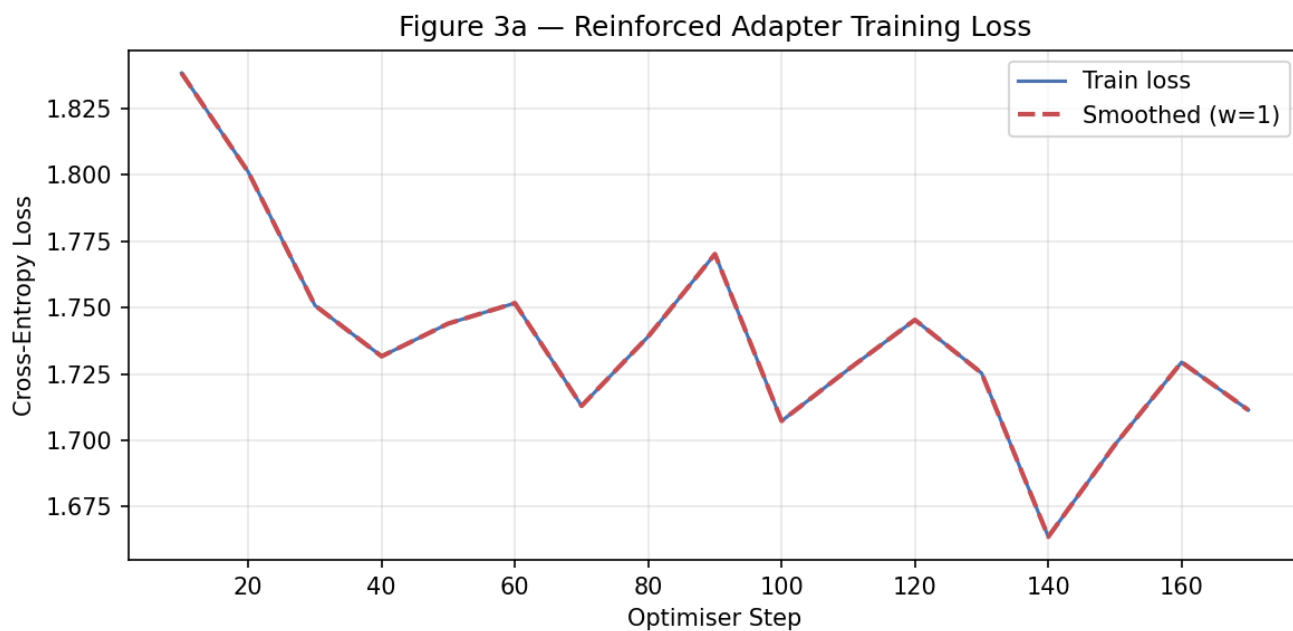


Figure 5. Adapter Loss for Dense unlearning results on Harry Potter.

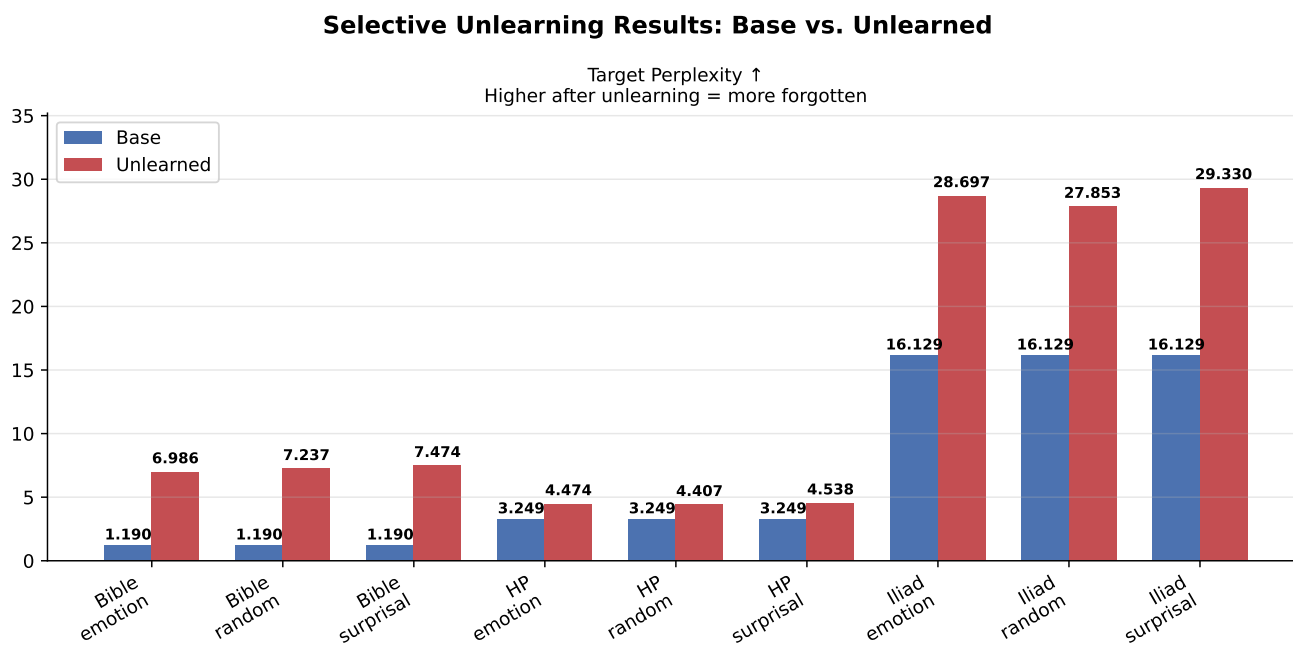


Figure 6. Sparse unlearning results for all corpus.

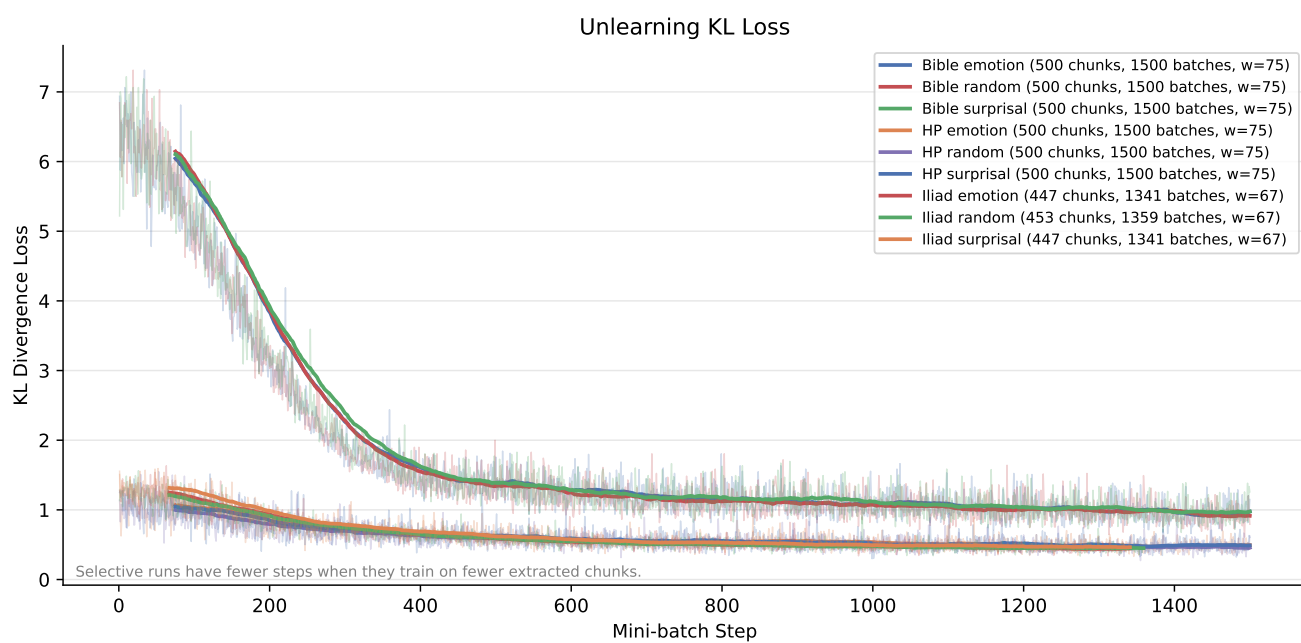


Figure 7. Training loss curves for sparse unlearning results on all corpus.